

CS534 — Written Homework Assignment 0
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Linear algebra

- (a) **Transpose and Associative Property, Positive Semi-definite matrices [2pt]** Define a matrix $B = bb^T$, where $b \in \mathbf{R}^{d \times 1}$ is a column vector that is not all-zero. Show that B is a positive semi-definite matrix.

[Hint: To show that B is positive semi-definite, we need to show that B is symmetric, and for any vector $x \in \mathbf{R}^{d \times 1}$, $x^T Bx \geq 0$. For the latter, try to get $x^T Bx$ to look like the product of two identical scalars. Note that $b^T x = (x^T b)^T$, that $a^T = a$ for scalar value a , and that matrix multiplication is associative.]

To show that B is positive semi-definite, we will need to verify two conditions:

1. B is symmetric.
2. For any vector $x \in \mathbf{R}^{d \times 1}$, we must show that $x^T Bx \geq 0$.

1. Symmetry:

$$B = bb^T$$

The transpose of B is:

$$B^T = (bb^T)^T = (b^T)^T b^T = bb^T = B$$

Since the transpose of a transpose is the original matrix so we can say that B is symmetric.

2. Positive semi-definiteness:

To show that for any $x \in \mathbf{R}^{d \times 1}$, $x^T Bx \geq 0$

$$\begin{aligned} x^T Bx &= x^T (bb^T)x \\ &= (x^T b)(b^T x) \\ &= (b^T x)^T (b^T x) = (b^T x)^2 \end{aligned}$$

Since the transpose of a transpose is the original matrix $(b^T x)^2$ is the square of a real number, it is always positive:

$$x^T Bx = (b^T x)^2 \geq 0$$

Therefore, $x^T Bx \geq 0$ for all x , so B is positive semi-definite.

- (b) **Solving systems of linear equations with matrix inverse. [2pt]** Consider the following set of linear equations:

$$\begin{cases} x_1 + x_2 - x_3 - x_4 = 1 \\ 2x_1 + 5x_2 - 7x_3 - 5x_4 = -2 \\ 2x_1 - x_2 + x_3 + 3x_4 = 4 \\ 5x_1 + 2x_2 - 4x_3 - 2x_4 = 6 \end{cases}$$

- (a) (1 pt) Please express the system of equations as $A\mathbf{x} = \mathbf{b}$ by specifying the matrix A and vector $\mathbf{b}$

The matrix A is:

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 2 & 5 & -7 & -5 \\ 2 & -1 & 1 & 3 \\ 5 & 2 & -4 & -2 \end{bmatrix}$$

The vector $\mathbf{x}$ is:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

The vector $\mathbf{b}$ is:

$$\mathbf{b} = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}$$

The equation can be expressed as:

$$A\mathbf{x} = \mathbf{b}$$

$$\begin{bmatrix} 1 & 1 & -1 & -1 \\ 2 & 5 & -7 & -5 \\ 2 & -1 & 1 & 3 \\ 5 & 2 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}$$

- (b) (1 pt) Solve for $A\mathbf{x} = \mathbf{b}$ by using the matrix inverse of A (you can use software to compute the inverse).

The above equation can also be written as:

$$\mathbf{x} = A^{-1}\mathbf{b}$$

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 2 & 5 & -7 & -5 \\ 2 & -1 & 1 & 3 \\ 5 & 2 & -4 & -2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}$$

The inverse of matrix A is:

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{6} & 0 & \frac{1}{6} \\ 2 & \frac{1}{6} & \frac{1}{2} & -\frac{2}{3} \\ \frac{7}{4} & -\frac{1}{4} & 0 & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$

Now, we multiply A^{-1} by $\mathbf{b}$ to get $\mathbf{x}$:

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{6} & 0 & \frac{1}{6} \\ 2 & \frac{1}{6} & \frac{1}{2} & -\frac{2}{3} \\ \frac{7}{4} & -\frac{1}{4} & 0 & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}$$

Now, we multiply A^{-1} by $\mathbf{b}$ to solve for $\mathbf{x}$:

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} \frac{11}{6} \\ -\frac{1}{3} \\ \frac{3}{4} \\ -\frac{1}{4} \end{bmatrix}$$

Thus, the solution to the system of equations is:

$$\mathbf{x} = \begin{bmatrix} \frac{11}{6} \\ -\frac{1}{3} \\ \frac{3}{4} \\ -\frac{1}{4} \end{bmatrix}$$

Vector Calculus

(a) **Derivatives.[2pt]**. Compute the derivative $f'(x)$ for

(a) (1 pts) the logistic (aka sigmoid) function $f(x) = \frac{1}{1+\exp(-x)}$

Given:

$$f(x) = \frac{1}{1 + e^{-x}}$$

For a fraction the derivative is given by:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{u'v - uv'}{v^2}$$

Here, let: - $u = 1$ (which has a derivative $u' = 0$) - $v = 1 + e^{-x}$

$$f'(x) = \frac{0 \cdot v - u \cdot v'}{v^2} = \frac{0 - 1 \cdot (-e^{-x})}{(1 + e^{-x})^2}$$

$$f'(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

We can write e^{-x} in terms of $f(x)$:

$$e^{-x} = \frac{1 - f(x)}{f(x)}$$

$$f'(x) = \frac{\frac{1-f(x)}{f(x)}}{\left(\frac{1}{f(x)}\right)^2}$$

$$f'(x) = f(x)(1 - f(x))$$

The derivative of the logistic function is:

$$\boxed{f'(x) = f(x)(1 - f(x))}$$

(b) (1 pts) $f(x) = \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$

The derivative of the exponential function is:

$$\frac{d}{du} \exp(u) = \exp(u)$$

Differentiating the inner function using the power rule:

$$\frac{du}{dx} = -\frac{1}{2\sigma^2} \cdot 2(x - \mu) = -\frac{(x - \mu)}{\sigma^2}$$

$$f'(x) = \frac{d}{dx} \left[\exp \left(-\frac{1}{2\sigma^2}(x - \mu)^2 \right) \right] = \exp \left(-\frac{1}{2\sigma^2}(x - \mu)^2 \right) \cdot \left(-\frac{(x - \mu)}{\sigma^2} \right)$$

$$\boxed{f'(x) = -\frac{(x - \mu)}{\sigma^2} \exp \left(-\frac{1}{2\sigma^2}(x - \mu)^2 \right)}$$

- (b) **Gradients.** [3pt] Compute the gradient $\nabla_{\mathbf{x}} f$ of the following functions. Please clearly specify the dimension of the gradient.

(a) (1pt)

$$f(z) = \log(1 + z), \quad z = \mathbf{x}^T \mathbf{x}, \quad \mathbf{x} \in R^D$$

The derivative of $f(z) = \log(1 + z)$ is:

$$\frac{df}{dz} = \frac{1}{1 + z}$$

Using the chain rule, we can represent the gradient $\nabla_{\mathbf{x}} f$ as:

$$\nabla_{\mathbf{x}} f = \frac{1}{1 + z} \cdot \nabla_{\mathbf{x}} z,$$

where $z = \mathbf{x}^T \mathbf{x}$

$$\nabla_{\mathbf{x}} z = \nabla_{\mathbf{x}} (\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}$$

Substituting the value of $\nabla_{\mathbf{x}} z = 2\mathbf{x}$ into the chain rule expression for $\nabla_{\mathbf{x}} f$:

$$\nabla_{\mathbf{x}} f = \frac{1}{1 + z} \cdot 2\mathbf{x}$$

$$\nabla_{\mathbf{x}} f = \frac{2\mathbf{x}}{1 + \mathbf{x}^T \mathbf{x}}$$

The gradient of the function is:

$$\nabla_{\mathbf{x}} f = \frac{2\mathbf{x}}{1 + \mathbf{x}^T \mathbf{x}} \in R^D$$

The dimension of the gradient is R^D because $\mathbf{x} \in R^D$

(b) (2pt)

$$f(z) = \exp\left(-\frac{1}{2}z\right)$$

$$z = g(\mathbf{y}) = \mathbf{y}^T S^{-1} \mathbf{y}$$

$$\mathbf{y} = h(\mathbf{x}) = \mathbf{x} - \mu$$

where $\mathbf{x}, \mu \in R^D, S \in R^{D \times D}$ is a symmetric matrix.

$$\frac{df}{dz} = -\frac{1}{2} \exp\left(-\frac{1}{2}z\right)$$

$$\nabla_{\mathbf{x}} f = \frac{df}{dz} \cdot \nabla_{\mathbf{x}} z,$$

(by using the chain rule) and $z = \mathbf{y}^T S^{-1} \mathbf{y}$ and $\mathbf{y} = \mathbf{x} - \mu$

The gradient of z with respect to $\mathbf{y}$ is:

$$\nabla_{\mathbf{y}} z = 2S^{-1} \mathbf{y}$$

Since $\mathbf{y} = \mathbf{x} - \mu$, the gradient of z with respect to $\mathbf{x}$ is:

$$\nabla_{\mathbf{x}} z = 2S^{-1}(\mathbf{x} - \mu)$$

Substituting this into the chain rule expression:

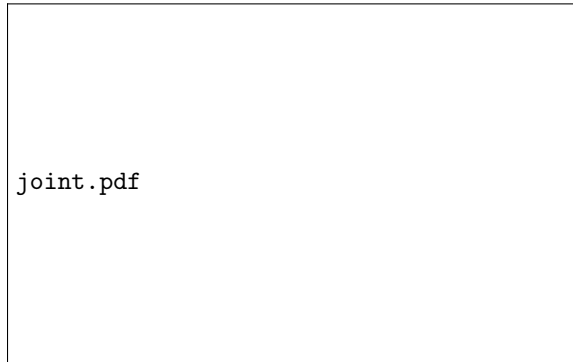
$$\nabla_{\mathbf{x}} f = -\frac{1}{2} \exp\left(-\frac{1}{2}z\right) \cdot 2S^{-1}(\mathbf{x} - \mu),$$

$$\nabla_{\mathbf{x}} f = -\exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T S^{-1}(\mathbf{x} - \mu)\right) S^{-1}(\mathbf{x} - \mu)$$

$$\nabla_{\mathbf{x}} f = -\exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T S^{-1}(\mathbf{x} - \mu)\right) S^{-1}(\mathbf{x} - \mu)$$

Probability

- (a) **Joint, Marginal, and Conditional Probabilities [2pt]** Consider two discrete random variables X and Y with the following joint distribution:



Please compute:

- (a) (1 pt) The marginal distributions $p(x)$ and $p(y)$

Marginal Distribution $p(x)$

$$p(x_1) = p(x_1, y_1) + p(x_1, y_2) + p(x_1, y_3) = 0.01 + 0.05 + 0.1 = 0.16$$

$$p(x_2) = p(x_2, y_1) + p(x_2, y_2) + p(x_2, y_3) = 0.02 + 0.1 + 0.05 = 0.17$$

$$p(x_3) = p(x_3, y_1) + p(x_3, y_2) + p(x_3, y_3) = 0.03 + 0.05 + 0.03 = 0.11$$

$$p(x_4) = p(x_4, y_1) + p(x_4, y_2) + p(x_4, y_3) = 0.1 + 0.07 + 0.05 = 0.22$$

$$p(x_5) = p(x_5, y_1) + p(x_5, y_2) + p(x_5, y_3) = 0.1 + 0.2 + 0.04 = 0.34$$

Marginal Distribution $p(y)$

$$p(y_1) = p(x_1, y_1) + p(x_2, y_1) + p(x_3, y_1) + p(x_4, y_1) + p(x_5, y_1) = 0.01 + 0.02 + 0.03 + 0.1 + 0.1 = 0.26$$

$$p(y_2) = p(x_1, y_2) + p(x_2, y_2) + p(x_3, y_2) + p(x_4, y_2) + p(x_5, y_2) = 0.05 + 0.1 + 0.05 + 0.07 + 0.2 = 0.47$$

$$p(y_3) = p(x_1, y_3) + p(x_2, y_3) + p(x_3, y_3) + p(x_4, y_3) + p(x_5, y_3) = 0.1 + 0.05 + 0.03 + 0.05 + 0.04 = 0.27$$

- (b) (1 pt) The Conditional distribution $p(x|Y = y_1)$ and $p(y|X = x_3)$

Conditional Distribution $p(x|Y = y_1)$

$$p(x_i|Y = y_1) = \frac{p(x_i, y_1)}{p(y_1)}$$

$$p(y_1) = \frac{26}{100}$$

$$p(x_1|Y = y_1) = \frac{1/100}{26/100} = \frac{1}{26}$$

$$p(x_2|Y = y_1) = \frac{2/100}{26/100} = \frac{1}{13}$$

$$p(x_3|Y = y_1) = \frac{3/100}{26/100} = \frac{3}{26}$$

$$p(x_4|Y = y_1) = p(x_5|Y = y_1) = \frac{10/100}{26/100} = \frac{5}{13}$$

Conditional Distribution $p(y|X = x_3)$

$$p(y_j|X = x_3) = \frac{p(x_3, y_j)}{p(x_3)}$$

$$p(x_3) = \frac{11}{100}$$

$$p(y_1|X = x_3) = \frac{3/100}{11/100} = \frac{3}{11}$$

$$p(y_2|X = x_3) = \frac{5/100}{11/100} = \frac{5}{11}$$

$$p(y_3|X = x_3) = \frac{3/100}{11/100} = \frac{3}{11}$$

- (b) **Conditional probabilities, Marginalization and Bayes Rule [5pt]** Consider two coins, one is fair and the other one has a 1/10 probability for the head. Now you randomly pick one of the coins and toss it twice. Answer the following questions.

- (a) (1pt) What is the probability that you picked the fair coin? What is the probability of the first toss being head?

Let assume x_1 be the outcome of the first toss and y denote the selected coin.

$$P(y = F) = P(y = B) = \frac{1}{2}$$

$$P(x_1 = H|y = F) = \frac{1}{2}$$

$$P(x_1 = H|y = B) = \frac{1}{10}$$

Probability of the first toss being head is:

$$P(x_1 = H) = P(x_1 = H, y = F) + P(x_1 = H, y = B)$$

$$= P(y = F)P(x_1 = H|y = F) + P(y = B)P(x_1 = H|y = B) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{10} = \frac{3}{10}$$

- (b) (2pts) If both tosses are heads, what is the probability that you have chosen the fair coin (Hint: you should apply Bayes Rule for this)?

Let's assume x_1, x_2 be the outputs of the first two tosses.

$$P(x_1 = H, x_2 = H|y = F) = \frac{1}{4}$$

$$P(x_1 = H, x_2 = H|y = B) = \frac{1}{100}$$

Using Bayes' theorem:

$$P(y = F|x_1 = H, x_2 = H) = \frac{P(x_1 = H, x_2 = H|y = F)P(y = F)}{P(x_1 = H, x_2 = H)}$$

$$\begin{aligned}
P(x_1 = H, x_2 = H) &= P(x_1 = H, x_2 = H|y = F)P(y = F) + P(x_1 = H, x_2 = H|y = B)P(y = B) \\
&= \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{100} \cdot \frac{1}{2} = \frac{13}{100} \\
P(y = F|x_1 = H, x_2 = H) &= \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{13}{100}} = \frac{25}{26}
\end{aligned}$$

- (c) (2pts) If both tosses are heads, what is the probability that the third coin toss will be head? (you should build on results of b)

Let's assume x_3 be the outcome of the third coin toss:

$$\begin{aligned}
P(x_3 = H|x_1 = H, x_2 = H) &= P(x_3 = H|y = F, x_1 = H, x_2 = H)P(y = F|x_1, x_2) + P(x_3 = H|y = B, x_1 = H, x_2 = H)P(y = B|x_1, x_2) \\
&= P(x_3 = H|y = F)P(y = F|x_1, x_2) + P(x_3 = H|y = B)P(y = B|x_1, x_2) \\
&= \frac{1}{2} \cdot \frac{25}{26} + \frac{1}{10} \cdot \frac{1}{26} = \frac{63}{130}
\end{aligned}$$

- (c) **Linearity of Expectation [2 pt]** A random variable x distributed according to a standard normal distribution (mean zero and unit variance) has the following probability density function (pdf):

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Using the properties of expectations, evaluate the following integral

$$\int_{-\infty}^{\infty} p(x)(ax^2 + bx + c)dx$$

[*Hint:* This is *not* a calculus question. The simple solution relies on the linearity of expectation and the provided mean/variance of $p(x)$.]

Given:

$$\int_{-\infty}^{\infty} p(x)(ax^2 + bx + c) dx$$

where $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ is the probability density function of a standard normal distribution
Here mean $\mu = 0$ and variance $\sigma^2 = 1$

Splitting the integration into three parts due to the linearity of the integral:

$$\int_{-\infty}^{\infty} p(x)(ax^2 + bx + c) dx = a \int_{-\infty}^{\infty} p(x)x^2 dx + b \int_{-\infty}^{\infty} p(x)x dx + c \int_{-\infty}^{\infty} p(x) dx$$

1.

$$\begin{aligned}
&\int_{-\infty}^{\infty} p(x)x^2 dx \\
E[X^2] &= \sigma^2 = 1
\end{aligned}$$

2.

$$\begin{aligned}
&\int_{-\infty}^{\infty} p(x)x dx \\
E[X] &= \mu = 0
\end{aligned}$$

3.

$$\int_{-\infty}^{\infty} p(x) dx$$

$$P(X) = 1$$

This is total probability

Substituting all these values in the given expression, we get:

$$\int_{-\infty}^{\infty} p(x)(ax^2 + bx + c) dx = a \cdot 1 + b \cdot 0 + c \cdot 1$$

$$= a + c$$

- (d) **Cumulative Density Functions / Calculus [2 pt]** X is a continuous random variable over the interval $[0,1]$, show that the following function p is a valid probability density function (PDF) and derive the corresponding cumulative density function (CDF).

$$p(x) = \begin{cases} 4x & 0 \leq x \leq 1/2 \\ -4x + 4 & 1/2 \leq x \leq 1 \end{cases}$$

[*Hint: Recall that a function is a valid PDF function if it integrates to 1: $\int_{-\infty}^{\infty} p(x) dx = 1$ And the cumulative density function (CDF) is defined as $C(x) = P(X \leq x)$ or the probability that a sample from p is less than x – which can be computed as $C(x) = \int_{-\infty}^x p(x) dx$ This is a calculus question. But the Pdf is a piece-wise linear function, hence it is straightforward.]*

To determine if the function $p(x)$ is a valid PDF and find its CDF, we need to check if

1. $p(x) \geq 0$ for all x
2. *The total area under the curve that is the integral of $p(x)$ across its range is equal to 1.*

PDF

$$p(x) = \begin{cases} 4x & \text{for } 0 \leq x < \frac{1}{2} \\ -4x + 4 & \text{for } \frac{1}{2} \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

For $0 \leq x < \frac{1}{2}$, $p(x) = 4x$, which is non-negative.

For $\frac{1}{2} \leq x \leq 1$, $p(x) = -4x + 4$ is also non-negative because:

At $x = 1$, $p(1) = 0$

At $x = \frac{1}{2}$, $p(\frac{1}{2}) = 2$

so $p(x) \geq 0$ for all x

$$\int_{-\infty}^{\infty} p(x) dx = \int_0^{\frac{1}{2}} 4x dx + \int_{\frac{1}{2}}^1 (-4x + 4) dx$$

1.

$$\int_0^{\frac{1}{2}} 4x dx = 4 \cdot \frac{x^2}{2} \Big|_0^{\frac{1}{2}} = 4 \cdot \frac{1}{8} = \frac{1}{2}$$

2.

$$\int_{\frac{1}{2}}^1 (-4x + 4) dx = [-2x^2 + 4x]_{\frac{1}{2}}^1 = (2) - \left(\frac{3}{2}\right) = \frac{1}{2}$$

Adding both:

$$\int_{-\infty}^{\infty} p(x) dx = \frac{1}{2} + \frac{1}{2} = 1$$

CDF

The CDF, $C(x)$, gives us the probability that the random variable X is less than or equal to x and is defined as:

$$C(x) = P(X \leq x) = \int_{-\infty}^x p(t) dt$$

1. For $x < 0$:

$$C(x) = 0$$

2. For $0 \leq x < \frac{1}{2}$:

$$C(x) = \int_0^x 4t dt = 2x^2$$

3. For $\frac{1}{2} \leq x \leq 1$:

$$C(x) = \int_0^{\frac{1}{2}} 4t dt + \int_{\frac{1}{2}}^x (-4t + 4) dt = \frac{1}{2} + (-2x^2 + 4x - 1) = -2x^2 + 4x$$

4. For $x > 1$:

$$C(x) = 1$$

$$C(x) = \begin{cases} 0 & \text{for } x < 0 \\ 2x^2 & \text{for } 0 \leq x < \frac{1}{2} \\ -2x^2 + 4x - 1 & \text{for } \frac{1}{2} \leq x \leq 1 \\ 1 & \text{for } x > 1 \end{cases}$$